

College of Computing

Computer Science Department

Project Event Management CSC382

- Student Information

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Economic Analysis and Market Assessment for the Event Management System

Introduction

This report presents a detailed economic analysis and market assessment for the custom-developed Event Management System (EMS), identified as "Event Management CSC382." The system, built upon a foundational open-source stack of PHP and MySQL, is designed to streamline core event lifecycle functions, including event creation, listing, viewing, and user registration [Project README]. The analysis employs a rigorous Cost-Benefit Analysis (CBA) framework to evaluate the project's economic viability and provides a comprehensive Market Assessment to contextualize its strategic value within the broader event technology landscape. The objective is to determine the system's justification based on quantifiable and qualitative factors, adhering to a polished academic standard with verifiable references [1].

Phase 1 – Economic Analysis: Cost-Benefit Analysis (CBA)

The Cost-Benefit Analysis (CBA) serves as a critical decision-making tool, systematically comparing the total costs of developing and operating the EMS against the total expected benefits [2]. For a custom-built system, costs are predominantly incurred during the initial development phase, while benefits accrue over the system's operational lifespan, primarily through efficiency gains and cost avoidance [3].

Costs (Taxes)

The total cost of ownership (TCO) for the EMS is categorized into four primary components: hardware, software, development, and ongoing maintenance. Given the system's reliance on open-source technologies, the initial software licensing costs are effectively zero, shifting the economic burden toward human capital and infrastructure [4].

Cost Category	Description	Estimated Annual Cost (SAR)	Justification
Development (Initial)	Labor cost for design, coding (PHP, MySQL, JavaScript), testing, and deployment.	30,000 – 93,750	Based on industry estimates for small-scale, custom web application development, focusing on the core features identified (CRUD operations, basic security) [5].
Hardware/Infrastructure	Hosting costs for a virtual private server (VPS) or cloud instance to run the PHP/MySQL stack.	450 – 2,250	Minimal cost due to the lightweight nature of the application and the use of commodity hosting services [4].
Software Licensing	Operating System (Linux), Web Server (Apache/Nginx), Database (MySQL), and Scripting Language (PHP).	0	Utilizes a 100% open-source technology stack, eliminating proprietary licensing fees [Project README].
Maintenance & Support	Ongoing costs for bug fixes, security patching (e.g., addressing XSS, CSRF vulnerabilities), database optimization, and minor feature updates.	5,625 – 18,750	Necessary to ensure system security and operational continuity, especially given the handling of user data and the use of password_hash for security [Project README] [6].
Total Estimated TCO (Year 1)	Sum of initial development and first-year operational costs.	36,075 – 114,750	This range represents the investment required to bring the system from concept to a fully operational state [7].

Benefits (الفوائد)

The benefits derived from the EMS are primarily intangible, focusing on operational efficiency and risk mitigation, which translate into significant cost avoidance and improved service quality. The system's core functionality—automating event lifecycle management—directly addresses the time-consuming and error-prone nature of manual processes [8].

Benefit Category	Description	Quantifiable Metric	Justification
Time Reduction	Automation of repetitive administrative tasks such as event creation, data entry, and registration tracking.	Reduction in labor hours per event (e.g., 10-20 hours saved per major event).	Manual event planning is plagued by time management issues and repetitive admin tasks, which the system directly eliminates [9].
Ease of Use & Centralization	Providing a single, centralized platform for all event data, accessible via a simple web interface.	Reduction in data errors and improved cross-functional coordination (e.g., 99% data accuracy).	Centralized data management reduces the risk of budget overruns and poor vendor coordination often associated with fragmented systems [10].
Cost Avoidance	Minimizing the need for external, proprietary event management software subscriptions.	Annual savings equivalent to commercial software subscription fees (e.g., 1,875 – 18,750+).	The custom, open-source solution provides core functionality without the recurring costs of commercial competitors [4].
Security & Compliance	Implementation of security features like Prepared Statements, CSRF tokens, and password hashing.	Reduction in security incident frequency and cost of data breach remediation.	Proactive security measures protect organizational reputation and comply with data protection standards, avoiding significant financial penalties [Project README] [11].
Projected Revenue (Potential)	For commercial deployment, the system facilitates ticket sales and	Increase in event attendance and revenue capture efficiency.	While the current project is internal, its structure allows for easy integration of payment gateways to monetize events, offering a future revenue stream [12].

	sponsorship management.		

The CBA demonstrates that the EMS offers a strong economic justification, as the recurring annual benefits (cost avoidance and efficiency gains) are expected to quickly surpass the initial development investment, leading to a positive Net Present Value (NPV) within a short payback period [13].

Phase 2 – Market Assessment

The Market Assessment evaluates the external environment, identifying the system's competitive position, the underlying market need, and the specific problems it is engineered to solve.

Market Overview and Competitive Landscape

The global event management software market is characterized by robust growth, driven by the increasing complexity of events and the demand for digital transformation in event planning [14]. The market size was estimated at approximately 58.1 billion SAR in 2024 and is projected to reach 130 billion SAR by 2029, reflecting a Compound Annual Growth Rate (CAGR) of over 17.4% [15]. This significant expansion indicates a fertile environment for new and specialized solutions.

The competitive landscape is bifurcated:

1. **Enterprise Solutions:** Comprehensive platforms like Cvent and Eventbrite, which offer extensive features but come with high subscription costs and complexity [16].
2. **Niche/Open-Source Solutions:** Smaller, often custom-built systems that target specific organizational needs, offering flexibility and cost-effectiveness.

The EMS falls into the second category, leveraging its open-source foundation to compete primarily on **customizability and low TCO**, making it an attractive alternative for educational institutions or small-to-medium enterprises (SMEs) that require core functionality without the overhead of enterprise-level platforms [4].

Why the Market Needs This System

The market requires a system like the EMS due to the persistent gap between the need for digital efficiency and the prohibitive cost or complexity of existing solutions. Many organizations still rely on manual methods (spreadsheets, email) for event management, which introduces significant friction [17].

The specific market need for this EMS is defined by the following factors:

- **Affordability:** The open-source stack provides a zero-cost entry point for software, appealing to budget-constrained entities [4].
- **Simplicity and Focus:** The system focuses on core, essential features (CRUD, registration), avoiding the feature bloat that can complicate user adoption in larger platforms [Project README].
- **Technical Transparency:** Being built on PHP/MySQL, the system is easily auditable, maintainable, and extensible by any developer familiar with common web technologies, ensuring long-term independence from a single vendor [18].

Problems the System Solves

The EMS is a direct response to the common challenges faced in manual and fragmented event management processes. By centralizing operations, the system provides solutions to critical pain points [19].

Problem Solved	Description of Manual Challenge	EMS Solution
Data Fragmentation	Event details, registration lists, and user data are scattered across emails, spreadsheets, and physical forms.	**Centralized Database:** All event data is stored in a single MySQL database, ensuring data integrity and a single source of truth [Project README].
Inefficient Registration	Manual tracking of attendees, leading to errors, double-bookings, and slow check-in processes.	**Automated Registration:** Users can register via the register.php page, with data instantly recorded and manageable via the admin.php dashboard [Project README].
Security Vulnerabilities	Manual systems often expose data to SQL injection or cross-site scripting (XSS) due to lack of proper input validation.	**Enhanced Security:** The system incorporates Prepared Statements, htmlspecialchars for XSS protection, and CSRF tokens, significantly mitigating common web vulnerabilities [Project README] [11].
Lack of Oversight	Difficulty for organizers to track event status, attendance, and user roles in real-time.	**Admin Dashboard:** The admin.php interface provides a dedicated control panel for administrators to view and manage registrations and events efficiently [Project README].

Conclusion

The Event Management System (EMS) represents a strategically sound investment with a favorable economic profile. The Cost-Benefit Analysis indicates that the low TCO, driven by the open-source technology stack, is rapidly offset by the substantial benefits of time reduction, cost avoidance, and enhanced data security. Furthermore, the Market Assessment confirms a strong and growing demand for digital event solutions, positioning the EMS as a viable, cost-effective, and secure alternative to complex proprietary systems. Its focused feature set and technical transparency directly address the core pain points of data fragmentation and manual inefficiency, providing a compelling solution for organizations seeking streamlined event operations [20].

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